

Hidden Display Structure Utilizing Micro-LEDs

Heumell Baek, SungMoo Kim, Gi-Hong Kim, JoonYoung Yang and SooYoung Yoon

LG Display Co., Ltd., Magokjungang 10-ro, Gangseo-gu, Seoul 07796, Korea

E-mail: cyberist@lgdisplay.com

Abstract

Recently smart surface technology has been actively studied for the automotive applications with hidden display systems emerging as a pivotal component in this field. This study proposes an innovative hidden display architecture that leverages the unique characteristics of micro-LEDs—particularly their compact light-emitting area—to achieve high efficiency and high resolution. Through proof-of-concept validation, we demonstrate the feasibility and performance advantages of this proposed structure, offering promising implications for next-generation automotive display systems.

Author Keywords

Smart Surfaces; Shy-tech; Hidden Display; Micro-LED

1. Introduction

Smart surface technology has recently emerged as a transformative approach to vehicle human-machine interfaces (HMIs), offering seamless integration of digital displays into automotive surfaces while maintaining aesthetic appeal and functional versatility. Within this rapidly evolving domain, hidden display systems have gained prominence as a pivotal component, enabling innovative visualization solutions that enhance both user experience and vehicle design [1, 2].



Figure 1. Example Image of Smart Surface.

Conventional hidden displays have been fabricated by attaching a hidden display film to the upper surface of the display as shown in Figure 2, which inevitably reduces both the display's transmittance and clarity. Although recent advancements have improved the transmittance of these films, a certain degree of performance degradation in the underlying display remains unavoidable due to the inherent structural limitations [1].

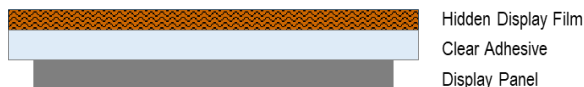


Figure 2. Simplified Structure of Conventional Hidden Display.

This study introduces an innovative hidden display architecture that capitalizes on the distinctive characteristics of micro-LED

technology—particularly their compact light-emitting area—to achieve high efficiency and high resolution. By addressing the inherent challenges associated with conventional display integration methods, our proposed structure enables superior performance.

Through comprehensive proof-of-concept (PoC) validation, we systematically demonstrate the technical feasibility and performance advantages of this novel architecture. Our experimental results reveal significant improvements in display efficiency and resolution capabilities, establishing a foundation for next-generation automotive display systems. The implications of this research extend beyond immediate automotive applications, potentially influencing broader developments in smart surface technology and human-machine interface design across multiple transportation and consumer electronics domains.

2. Proposal of New Hidden Display Structure

To enhance the optical efficiency of hidden displays, we focused on a key characteristic of micro-LEDs: their small light-emitting area. We redesigned the overall stacked structure of the hidden display to minimize absorption along the optical path of the micro-LED light source. To this end, we developed a method that decouples the two primary functional components of the hidden display film—the background color and the foreground pattern—by integrating the background color directly into the display panel and fabricating the foreground pattern on a separate film for subsequent attachment, as illustrated in Figure 3.

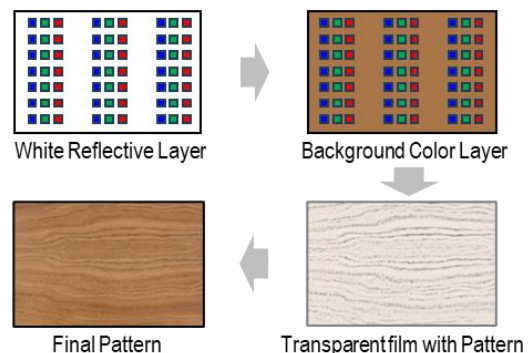
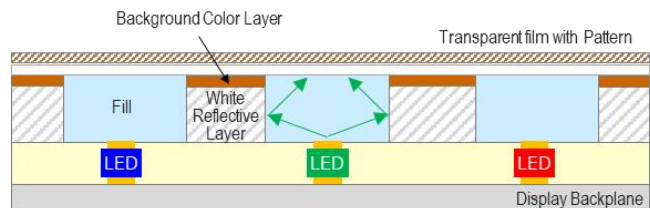


Figure 3. Proposed Hidden Display Structure utilizing Micro-LEDs.

3. Proof of Concept

Preparation of Micro-LED panel: We selected a 10.4-inch micro-LED panel with a resolution of 200 ppi to validate the high-efficiency hidden display concept. Each pixel of this panel incorporates two $\Phi 10\mu\text{m}$ blue vertical micro-LED chips. Following the completion of the LED electrode interconnection process, a white reflective layer was deposited onto the panel. Using a conventional photolithography process, this reflective layer was patterned while considering the positions of the LED chips (see Figure 4(b)). The white reflective layer serves two critical functions: first, it enhances the visibility and accurate rendering of colors, analogous to the function of white paper in printed media, and second, it improves the light extraction efficiency of the micro-LED chips, thereby increasing overall luminous output.

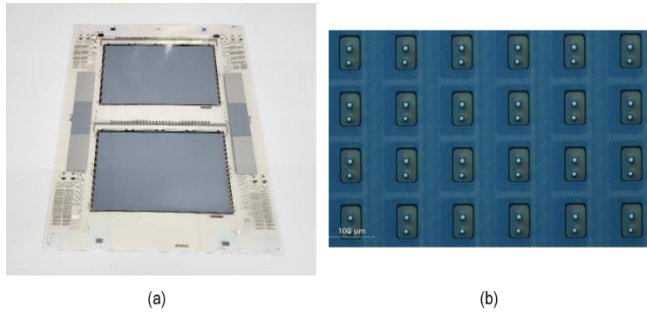


Figure 4. (a) Photograph of the panel glass after the formation of the white reflective layer. (b) Microscopic image of the corresponding pixel array. The photograph was captured under yellow lighting conditions, which unfortunately resulted in an inaccurate representation of the actual color.

During the PoC process, instead of newly developing a dedicated color resin for achieving a specific hue, we opted to create a brown-like color by mixing existing red and green resins. By adjusting the ratio of red to green, we were able to achieve a color closely resembling brown (see Figure 5). In the future, it may be feasible to develop and utilize more suitable color resins as needed, depending on specific application requirements. The background color was successfully formed on top of the white reflective layer by coating the mixed color resin. This background color layer was also patterned by photolithography process as well.

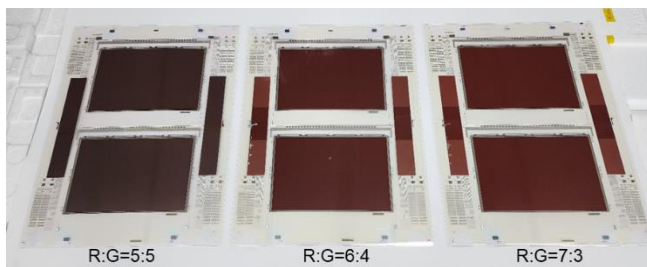


Figure 5. Photograph of the panel glass after the formation of the background color layer.

Preparation of transparent film with pattern: To prepare a patterned transparent film, inkjet printer-compatible overhead projector (OHP) film and waterslide decal paper were utilized. A

wood grain pattern with the background removed was printed using a conventional inkjet printer onto these materials to produce the transparent film.

Figure 6 shows images of transparent films with different patterns placed over panels that have background colors. As demonstrated, separating the background color and the foreground pattern during fabrication, and combining them in the final step allows achieving the desired effect, as intended in this study.



Figure 6. Images of various combinations of printed transparent film with pattern on top of display panel with background color.

Final Display Evaluation: The fabricated PoC display was combined with a driving circuit to test actual performance of displayed image. It was confirmed that the foreground patterns of transparent pattern film did not much interfere when the image was activated by the bright micro-LEDs, resulting in clear and bright image display as shown in Figure 7. The relative efficiency of final PoC panel showed 96% of that of reference panel at the same power consumptions (See Table 1).



Figure 7. Display image of turned-on PoC panel.

Table 1. Relative Panel Efficiency Comparisons.

Ref. Panel	PoC Panel	
	background color only	w/ transparent pattern film
100%	110%	96%

The transparent pattern film based on OHP film could not be physically attached, resulting in an air gap between the film and the display, which somewhat degraded the image quality. However, when using a waterslide decal paper film, it was possible to attach it directly to the display without an air gap, thereby achieving a clearer image quality. In the future, additional research on improving image quality through collaboration with specialized film companies or the use of adhesive layers will be necessary.

4. Discussion and Conclusion

Through the PoC process and results of this study, it was confirmed that the proposed micro-LED display structure reduces light absorption by the hidden display film layer, leading to higher efficiency while maintaining the aesthetic appeal required for hidden displays when the power is off. The findings of this research are significant because they improve power efficiency—a known weakness of existing hidden displays—and suggest the potential for a display structure that is better suited for high-resolution image representation.

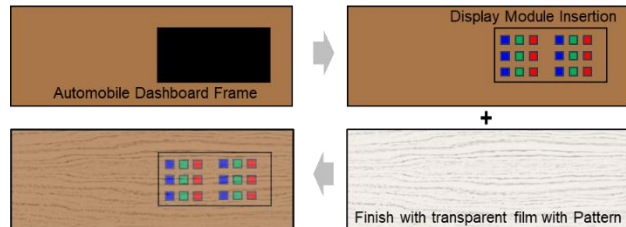


Figure 8. Dashboard application procedure concept.

To apply this technology to an actual car dashboard, the topmost film must be attached over a wider area than the display, as illustrated in Figure 8. Although further research and development are required for practical implementation, it is expected that collaboration among related companies will enable successful realization.

Furthermore, this technology is not limited to hidden displays; it also allows for the implementation of various off-state colors of display devices, rather than being confined to a black color when the display is turned off. The technique of directly applying the background color to the display panel is easy to implement as described in PoC section and allows for displays in various colors depending on the resin used. Figure 9 shows examples of display panels using White/Blue color resin.



Figure 9. Display with different background color.

5. Acknowledgements

This work was supported by the Technology Innovation Program(20022445) funded by the Ministry of Trade, Industry&Energy(MOTIE, Korea). The authors also thank the personnel of ChemE co., ltd. who provided the white reflective layer and color layer materials for this study.

6. References

1. Kai Hohmann. Novel Automotive Display Experiences beyond Large Display Areas, 50-1, SID2025. Available from: <https://doi.org/10.1002/sdtp.18251>
2. Juergen Baethis and Kai Hohmann, Shy Tech Displays High resolution Displays Hidden Behind Decorative Surfaces, 61-2, SID2022. Available from: <https://doi.org/10.1002/sdtp.15612>